Producers, consumers and decomposers

All the organisms in an ecosystem can be grouped into one of three types, depending on what they eat.

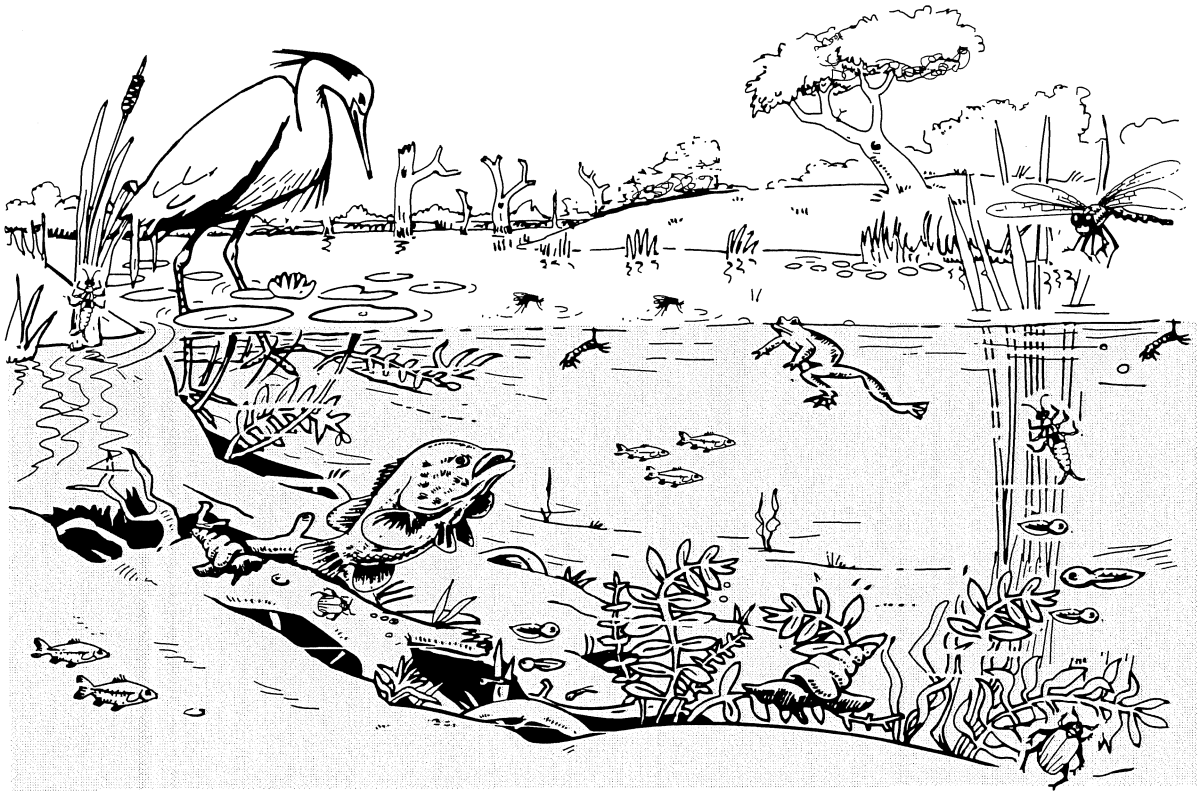
- **Producers** are living things that make (or produce) their own food from inorganic (non-living) ingredients. Examples of producers are green plants.
- **Consumers** are organisms that cannot make their own food. They either eat or use food products made by producers or other living things that eat producers. Examples of consumers are insects and snakes.
- **Decomposers** break down organic matter (dead plants and animals) by making it rot or decay, releasing nutrients, and then absorb the nutrients as food. Examples of decomposers are bacteria and fungi. Many decomposers are microscopic organisms, called microorganisms.

☞ **1 a)** In the diagram below, colour any producers GREEN and any consumers RED.

b) Can you see any decomposers? Why or why not?

_____

_____



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What is a consumer?

BACKGROUND

mammals (say 'ma-mels') – warm-blooded animals with backbones that feed their young with milk

producers (say 'pro-dews-ers') – living things that make their own food from non-living ingredients

Consumers are types of organisms that eat or use food products made by producers or that eat other consumers. Large land mammals, birds and reptiles are consumers.

There are three main types of consumers:

- **herbivores** (consumers that eat plants)
- **carnivores** (consumers that eat animals, or meat)
- **omnivores** (consumers that eat both plants and animals).

Humans are omnivores. We eat all sorts of foods produced by plants, and we also eat other animals that eat plants and animals.

☞ **1** List six types of living things that are consumers.

_____ _____

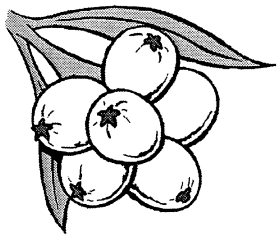
_____ _____

_____ _____

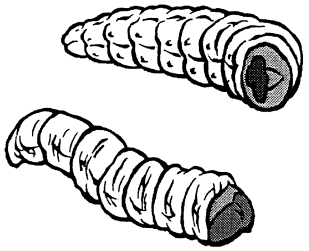
2 Decide whether these things are producers or consumers. Circle your response for each one.



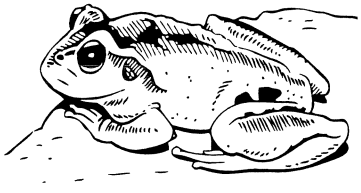
a producer OR consumer



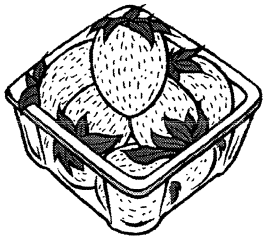
b producer OR consumer



c producer OR consumer



d producer OR consumer



e producer OR consumer

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What is a consumer? CONT'D

Most carnivores are **predators**. Predators hunt, kill and eat other living things. The living thing that a predator hunts is called its **prey**. For example, the African lion is a fierce predator that needs meat to survive. Its prey includes wildebeest, zebras and giraffes.

3 a) Which of the living things in the box are predators? Underline your choices.

b) Which of the living things in the box are prey for different types of predators? Circle your choices.

tiger	caterpillar	snail	zebra
brown bear	mouse	rabbit	dingo

c) Look at the living things you have underlined. List each one below and write whether that predator is a carnivore or omnivore.

Predator: _____ Carnivore or omnivore? _____

Predator: _____ Carnivore or omnivore? _____

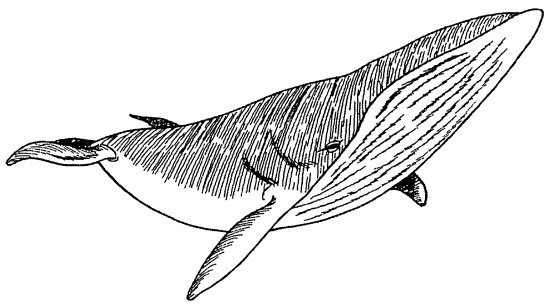
Predator: _____ Carnivore or omnivore? _____

4 a) Which of these animals have no natural predators to fear in their ecosystems? Circle your choices.

blue whale	lizard	prawn
rabbit	crocodile	lion

b) Look at the animals you have circled. What animal hunts and kills these animals, but is not considered a natural predator?

_____



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What is a decomposer?

BACKGROUND

nutrients (say 'nu-tree-ants') – substances that give animals and plants the energy essential for growth and life

A decomposer is a living thing that works to break down **organic** materials into smaller parts. For example, a decomposer breaks down fruit on a forest floor by making it rot or decay. The decomposer absorbs the nutrients from the remains and also releases some nutrients into the soil.

Fungi and moulds are decomposers. They decompose plant waste, including vegetables. Some small organisms from the animal kingdom are also decomposers, including worms, slugs, millipedes and termites, which break down logs and sticks. Other decomposers, like maggots and flies, help to break down dead animal carcasses (bodies). Dung (animal droppings) is broken down by dung flies and beetles. Microorganisms, including various bacteria, break down all kinds of substances, including milk, eggs and meat products.

☞ **1** Circle six things that are classified as decomposers.

termite	bacteria	mould	toadstool
maggot	rabbit	worm	eagle

2 Write these words into the spaces below. The lines break them into syllables.

decompose ___ ___ / ___ ___ ___ / ___ ___ ___ ___

decomposers ___ ___ / ___ ___ ___ / ___ ___ ___ / ___ ___ ___

decomposition ___ ___ / ___ ___ ___ / ___ ___ ___ / ___ / ___ ___ ___ ___

3 Complete this passage by filling in the missing vowels (a, e, i, o, u).

D ___ c ___ mp ___ s ___ rs break down dead ___ rg ___ n ___ sms into sm ___ ller parts. Leaf litter, animal d ___ ng, bodies of d ___ ___ d creatures and dead pl ___ nts all d ___ c ___ mp ___ s ___. After decomposition, the n ___ trients that were contained in the decayed organisms return to the s ___ ___ l. In this way, they are r ___ cycl ___ d. Nutrients in the soil are then taken up into y ___ ___ ng plants and ___ n ___ mals eat the plants. The cycle of life and death begins again.

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Food chains

BACKGROUND

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

In an ecosystem, one organism depends on another to survive. Producers make food, which is then eaten by consumers, which is then broken down by decomposers. Consumers and decomposers turn this food into the energy they need to survive. This creates what scientists call a **food chain**.

A food chain can be represented using pictures or words, but it always needs arrows. The arrows show what is eaten by what and the **flow of energy** along the food chain. They point away from the thing that is being eaten. For example:

plant → grasshopper → snake → eagle

Sometimes food chains show decomposers. Sometimes they show just producers and consumers, or just consumers.

☞ **1** Read these statements and then write a food chain for each. Two examples have been done for you. Hint: Make sure your organisms are in the correct order. The correct order may not be the order in which they appear in the statement. Check your answers by replacing each arrow in the food chain with 'is eaten by'.

a) A scorpion is eaten by a meerkat, which is eaten by a cobra, which is eaten by a vulture.

scorpion → meerkat → cobra → vulture

b) A tiger eats a deer, which eats a plant.

plant → deer → tiger

c) A ferret eats a rabbit, which eats grass.

_____

d) A brown bear eats a salmon, which eats plankton.

_____

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Food chains CONT'D

e) Seaweed is eaten by a small fish, which is eaten by a tuna, which is eaten by a cormorant.

_____

f) A lynx eats a raccoon, which eats a rat, which eats an insect.

_____

g) A wolf eats a woodpecker, which eats a frog, which eats a fly.

_____

h) Grain is eaten by a fieldmouse, which is eaten by a weasel, which is eaten by a cougar.

_____

i) A shark eats a jewfish, which eats a whiting, which eats a prawn, which eats a worm.

_____

j) Fruit is eaten by a grub, which is eaten by a monkey, which is eaten by a hyena.

_____

k) A crab eats a sea snail, which eats a starfish, which eats a mussel, which eats plankton.

_____

Food chains are not the best way to describe the relationships in most ecosystems. They are too simple. Ecosystems usually involve many species that are connected in all sorts of ways.

For example, this food chain is from an African savannah: grass → zebra → lion. There are many savannah animals that consume grass, and many that are eaten by lions. Birds also eat the remains of zebras, and decomposers like bacteria, flies and microbes eat them as well. The best way to show all of these relationships is using a **food web**.

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Food webs

BACKGROUND

habitat (say 'hab-i-tat') – the particular place where a group of living things live

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

Instead of drawing or writing a food chain, we can better show the many relationships between living things in an ecosystem by drawing a food web.

There is often more than one species that eats another in an ecosystem. Food webs are used to describe the diets of animals that have a variety of food sources, and they are also used to show all the relationships between creatures in a particular ecosystem.

☞ **1** Look at the food web pictured and answer the following questions.

a) What eats the frog?

_____

b) What is eaten by the chicken?

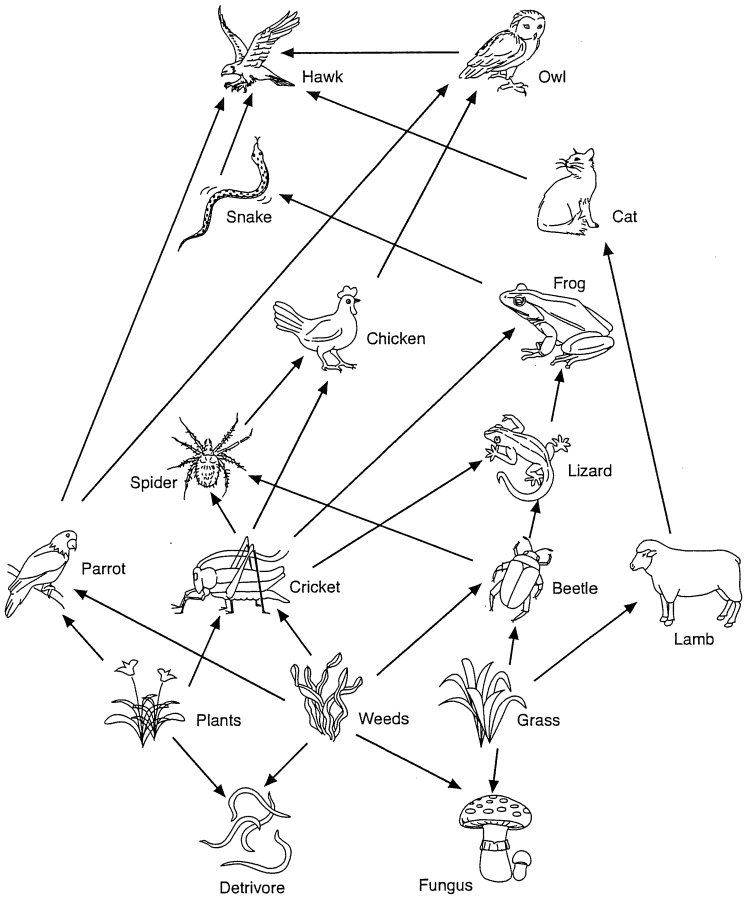
_____

c) What is consumed by the cricket?

_____

d) What kind of ecosystem do you think this food web comes from?

_____



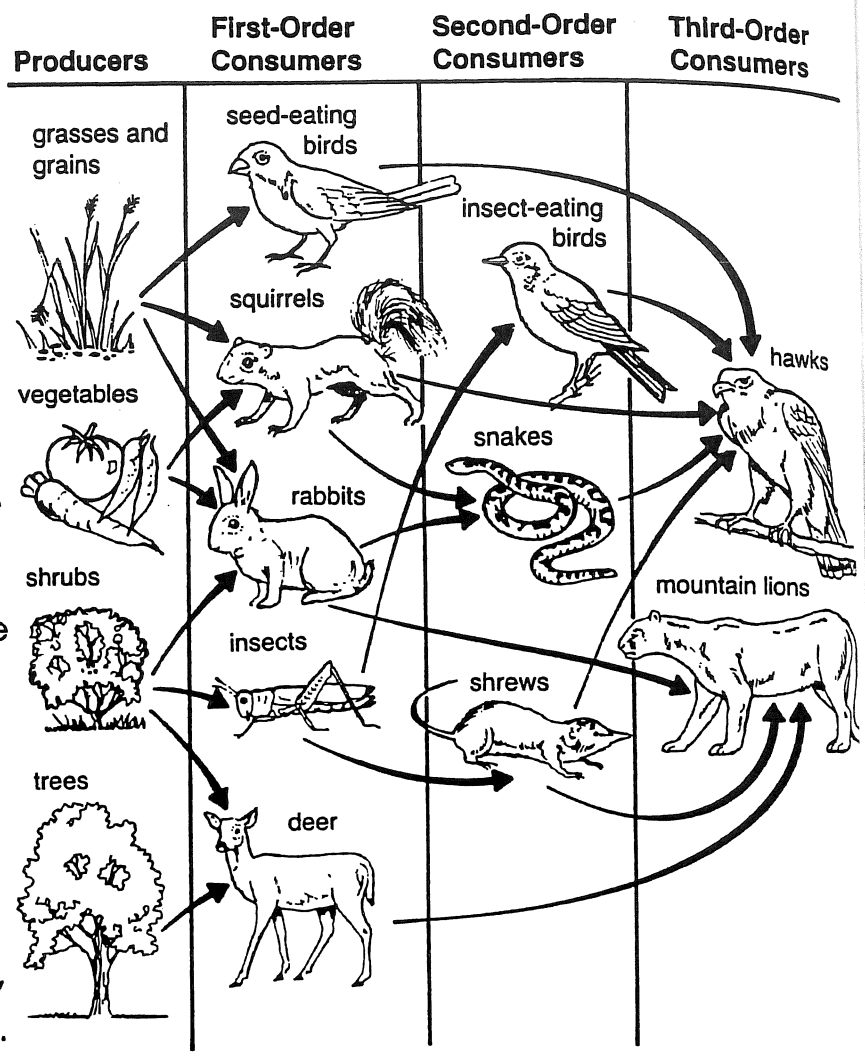
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ECOLOGICAL RELATIONSHIPS

Name _____

Food Webs

Use the food web below to answer questions 1-5.



1. When the hawk is the third-order consumer, the number of second-order consumers is _____.
2. The food chain that includes insect-eating birds is ____ _____.
3. The animal that consumes the largest number of different types of first-order and second-order consumers is the _____.
4. All the animals that are herbivores are _____ consumers.
5. If there were no snakes in this food web, the squirrels and rabbits could still be eaten by the _____.

Ecological Terms

Fill in the blanks with the words from the list below.

community		*decomposers*	*ecosystem*	*habitat*
herbivore	*niche*	*omnivore*	*population*	*scavenger*

6. A group of independent organisms in a particular environment is a(an) _____.
7. Organisms that break down complex compounds and dead tissue are _____.
8. All the living and nonliving things in a selected area form a(an) _____.
9. The particular environment to which a particular species is adapted is its _____.
10. The special role and place of an organism within its habitat is its _____.
11. An animal that eats both plants and animals is a(an) _____.
12. Members of a single species that occupy a common area form a(an) _____.
13. Animals that feed only on dead organisms are _____.

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